

Percentages



1 "out of 100"

2

a 100

b 53

3

a

i 71

ii 71%

iii $\frac{71}{100}$

b

i 50

ii 50%

iii $\frac{1}{2}$

Converting from percentages

4

a $\frac{17}{25}$

b $\frac{13}{100}$

c $\frac{11}{20}$

d $2\frac{3}{4}$

e $\frac{37}{100}$

f $\frac{47}{50}$

g $\frac{9}{2}$

h $\frac{1}{40}$

5

a 0.54

b 0.6

c 2.7

d 3.03

e 0.76

f 5

g 1.82

h 68.97

Converting to percentages

6

a

i $\frac{50}{100}$

b

i $\frac{3}{100}$

c

i $\frac{12.5}{100}$

d

i $\frac{22.5}{100}$

ii 50%

ii 3%

ii 12.5%

ii 22.5%

7

a 83%

b 180%

c 42%

d 65%

e 62%

f 60%

g 370%

h 218%

8

- a No
b 67%



9

- a 13% b $\frac{4}{25} = 16\%$ c $\frac{63}{90} = 70\%$ d $\frac{1}{3}$

10

- a 24% b 110% c 85.7% d 800%
e 171% f 2400% g 1304% h 4.1%

Comparing different values

11 Multiplying by 100% introduces the % symbol without changing the number's value.

12 Dividing by 100% removes the % symbol without changing the number's value.

13

Fraction	Decimal	Percentage
$\frac{11}{100}$	0.11	11%
$1\frac{83}{100}$	1.83	183%
$\frac{5}{8}$	0.625	$62\frac{1}{2}\%$

14

- a $\frac{4}{5}$ and 0.8 b 0.475 and $\frac{19}{40}$

15

- a
i 50%, 100%, 30%
ii 0.3, $\frac{5}{10}$, 100%
- b
i 25%, 60%, 30%
ii $\frac{1}{4}$, 0.3, 60%
- c
i 37.5%, 20%, 80%
ii 20%, $\frac{3}{8}$, 0.8
- d
i 20%, 45%, 27%
ii $\frac{3}{15}$, 0.27, 45%